

Unit 3

Solving systems of linear equations.

Unit Goals

- Be able to explain both the “row picture” and “column picture” of a linear system and its solutions.
- Be able to translate between augmented matrices and systems of equations.
- Be able to **apply** Gaussian elimination to solve a system of linear equations with a unique solution.
- Be able to identify when a linear system has either no solution or many solutions.
- Be able to express the solution set to a linear system with many solutions in parametric form.
- Be able solve real-world problems with linear models using the tools of Gaussian elimination.

Unit 3.1

**The geometry of linear systems
and their solutions.**

What is a linear model?

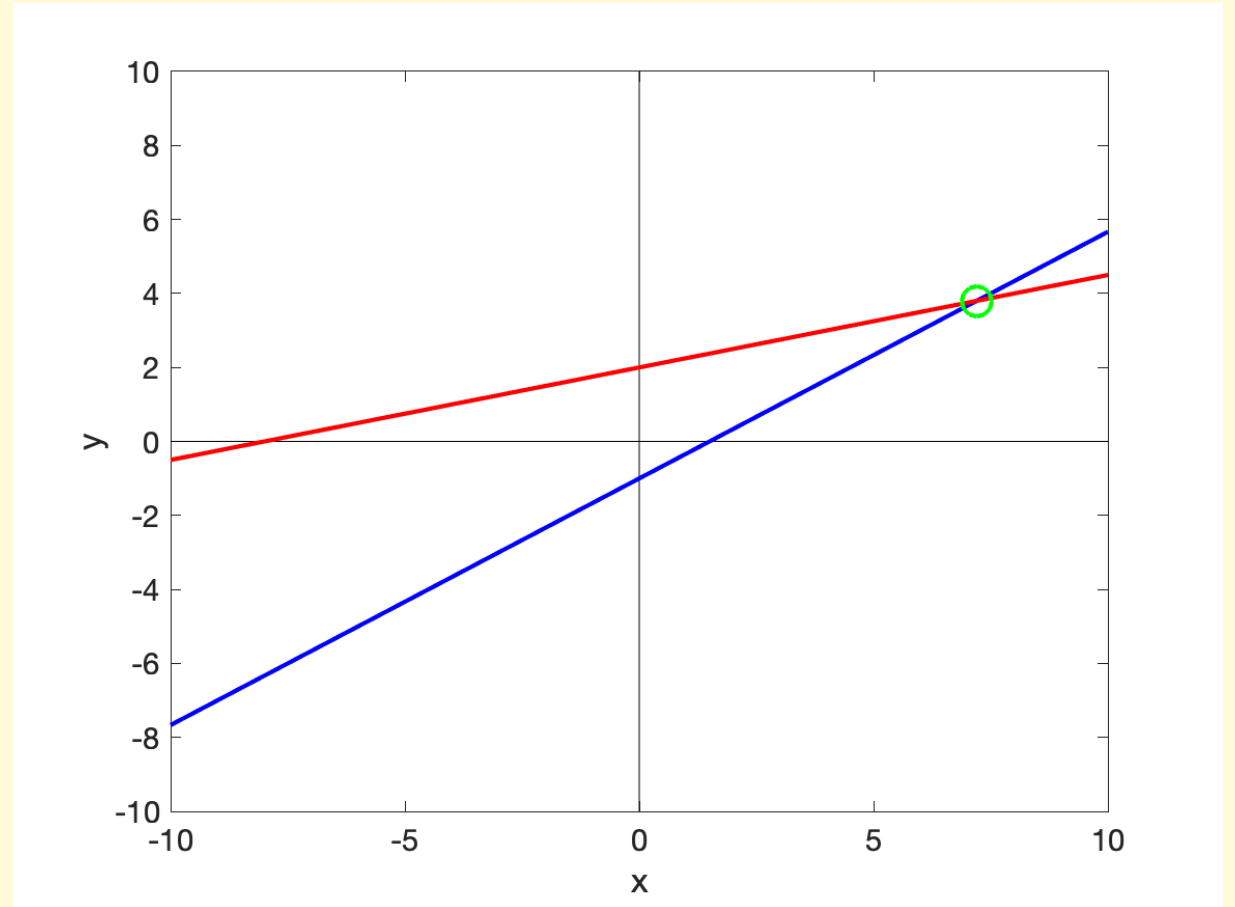
- A linear model is a mathematical model described by a linear system of equations.
- A system of equations is a set of equations that use the same variables.
- **The variables have the same meaning, and must take on the same value, in each equation in the system.**
- A linear system of equations is simply a system in which every equation is linear (all terms are of the form of constant $\times$ variable (ax) or constant (a)).

The geometry of linear systems.

- The general form of a linear equation in 2-D is:

$$a_1x_1 + a_2x_2 = b \rightarrow x_2 = -\frac{a_1}{a_2}x_1 + \frac{b}{a_2}, a_2 \neq 0$$

- This is of the form $y = mx + b$, the equation for a straight line with slope m and y-intercept b .
- A system of linear equations in 2-D is thus a set of lines.
- Solutions to the system are points where the lines intersect.



The geometry of linear systems.

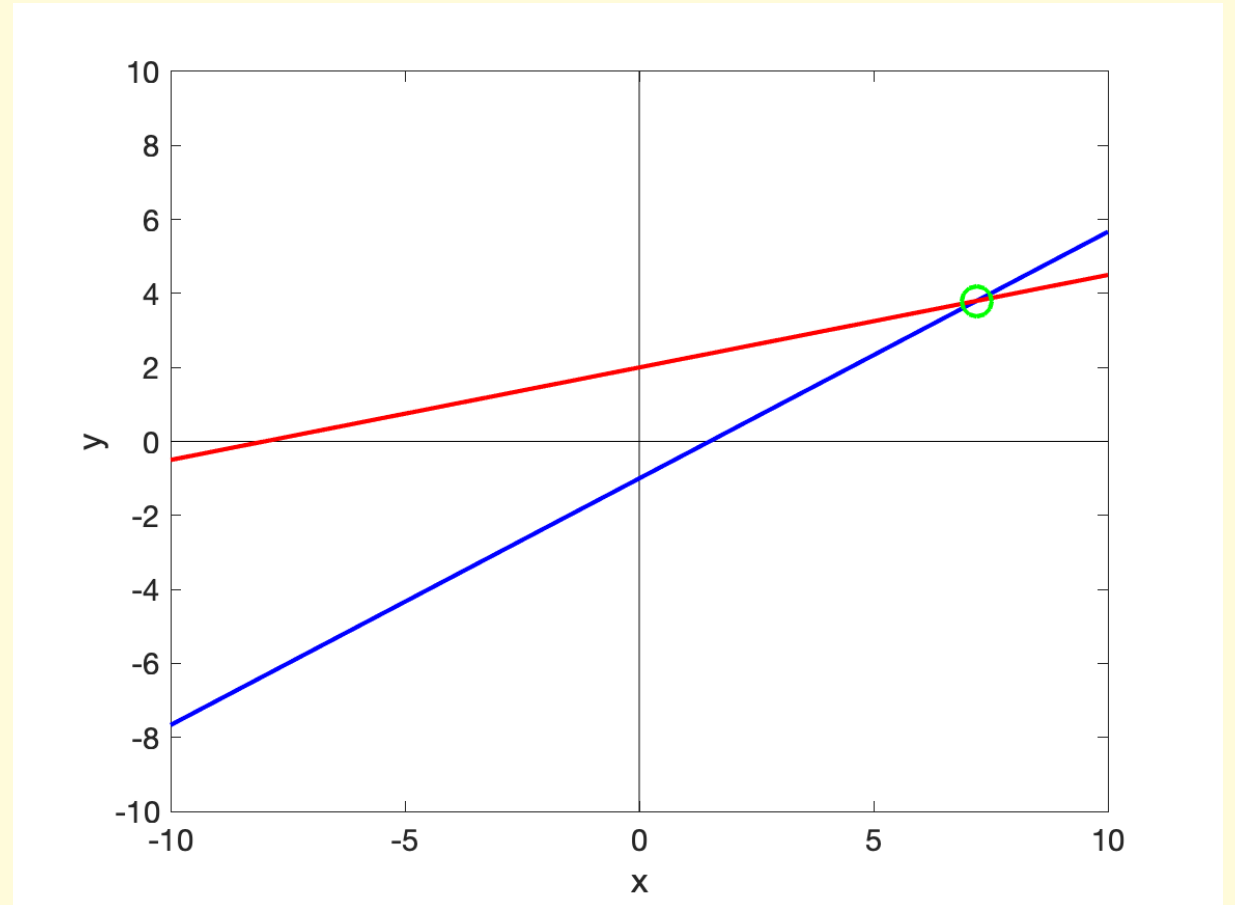
- Two lines in 2-D usually intersect as a single point.

$$2x - 3y = 3 \rightarrow y = \frac{2}{3}x - 1$$

$$x - 4y = -8 \rightarrow y = \frac{1}{4}x + 2$$

- The system has one unique solution.

Solution: $(x, y) = (7.2, 3.8)$



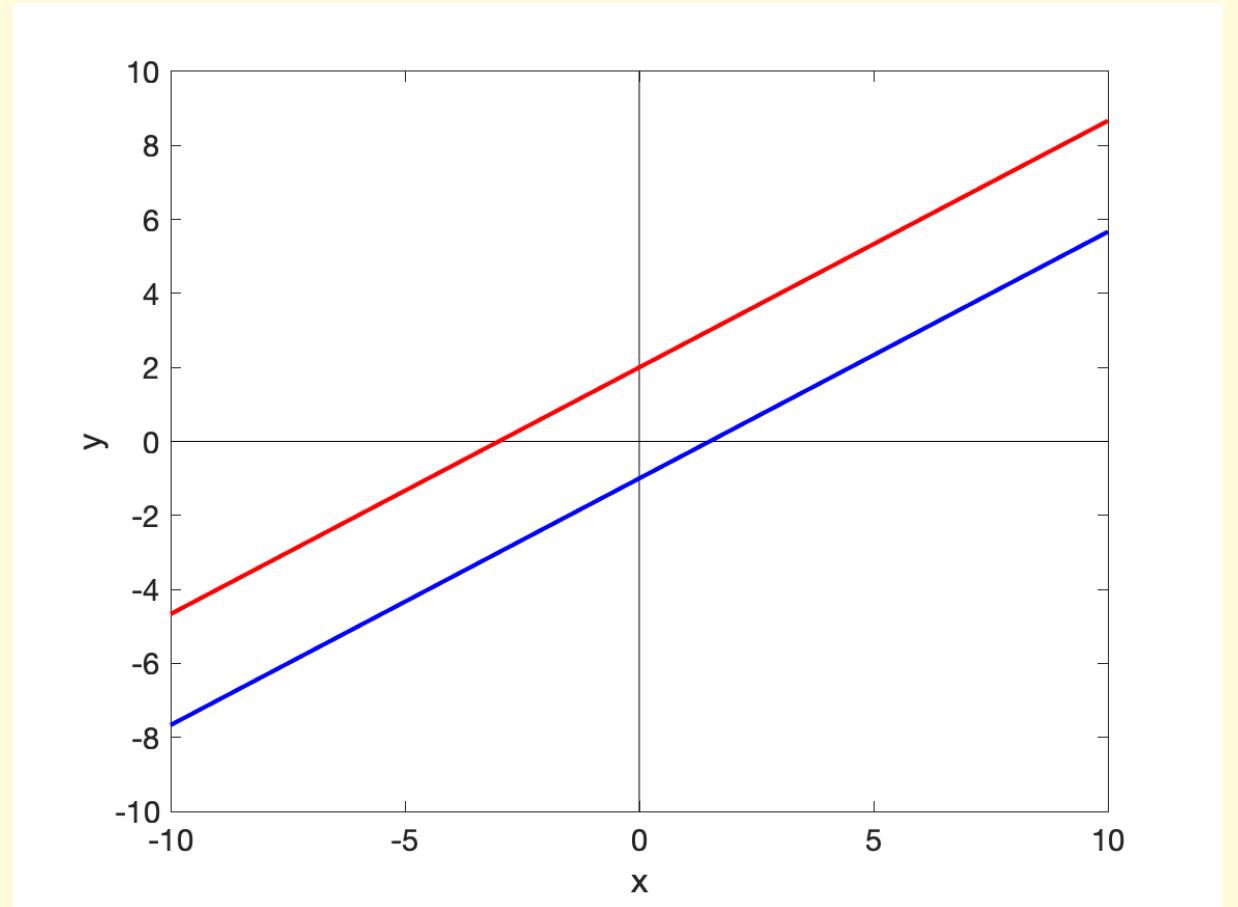
The geometry of linear systems.

- Parallel lines never intersect:

$$2x - 3y = 3 \rightarrow y = \frac{2}{3}x - 1$$

$$2x - 3y = -6 \rightarrow y = \frac{2}{3}x + 2$$

- **The system has no solution.**



The geometry of linear systems.

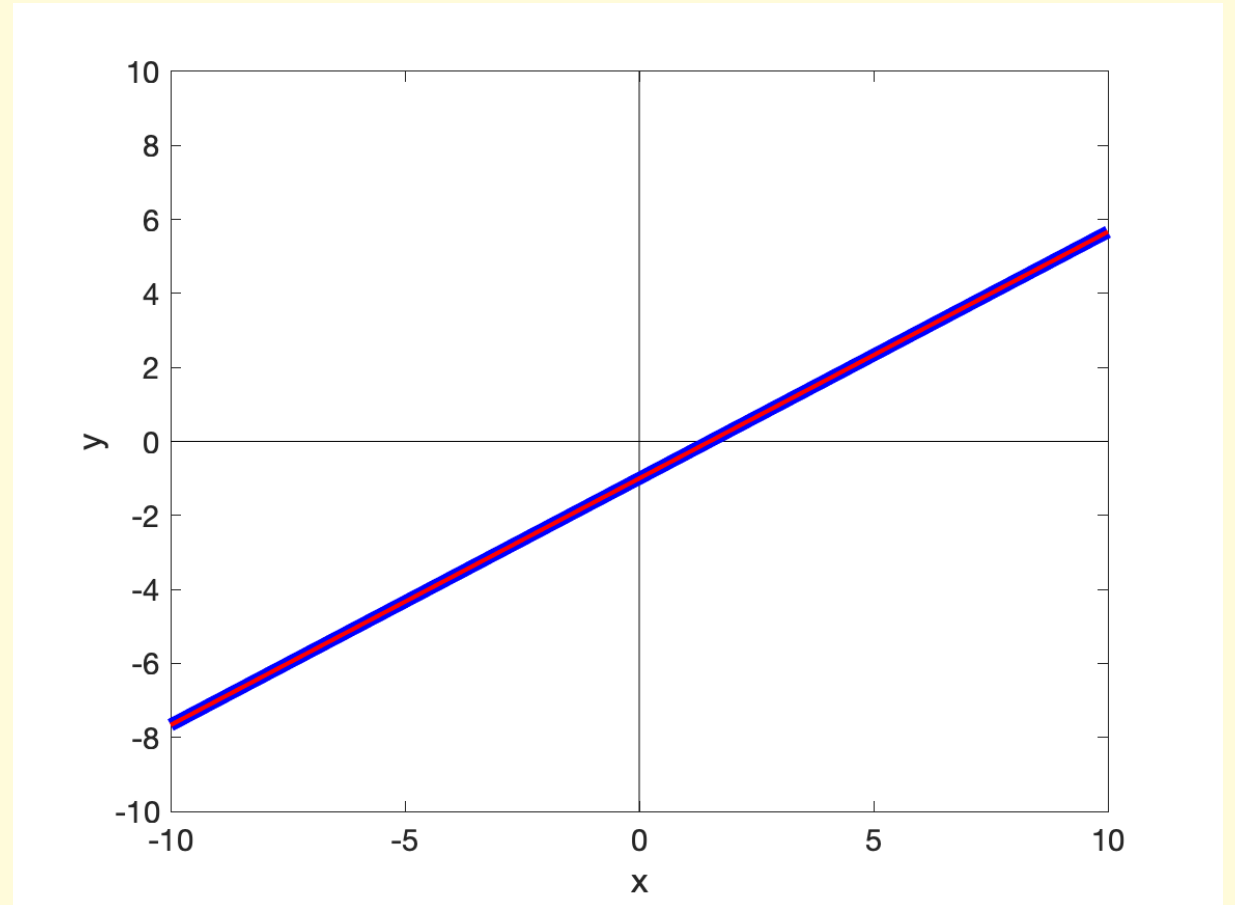
- Overlapping lines intersect everywhere on the line:

$$2x - 3y = 3 \rightarrow y = \frac{2}{3}x - 1$$

$$4x - 6y = 6 \rightarrow y = \frac{2}{3}x - 1$$

- The system has many solutions.
 - There are an infinite number of solutions that we can express in 'parametric' form.

Solution: $(x, y) = \left(t, \frac{2t}{3} - 1\right), \forall t$



The geometry of linear systems.

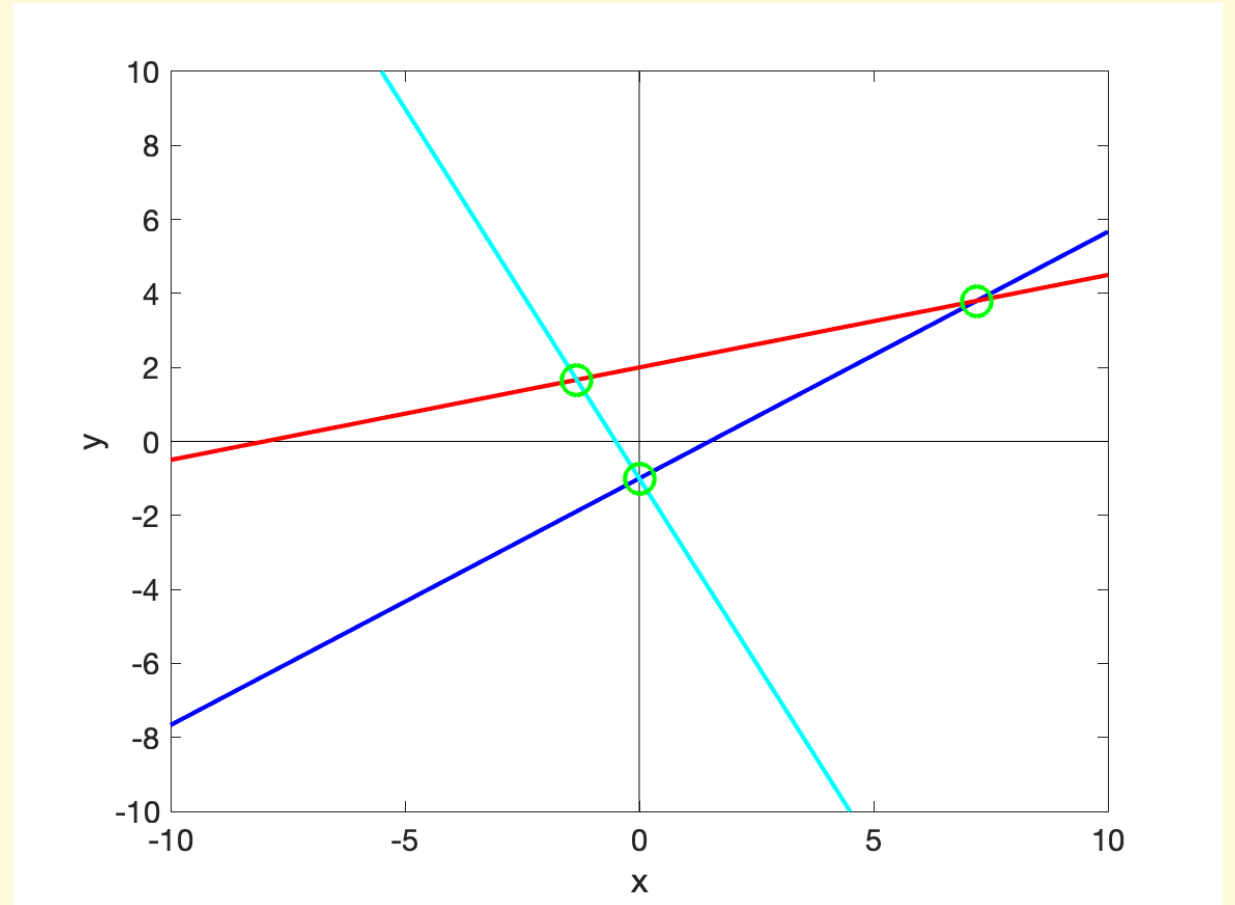
- Three lines in 2-D usually do not all intersect at the same point.

$$2x - 3y = 3 \rightarrow y = \frac{2}{3}x - 1$$

$$x - 4y = -8 \rightarrow y = \frac{1}{4}x + 2$$

$$2x + y = -1 \rightarrow y = -2x - 1$$

- Three intersection points, but all only involve two lines.
- **The system has no solution.**



The geometry of linear systems.

- In special circumstances, 3 lines in 2-D can all intersect at the same point.

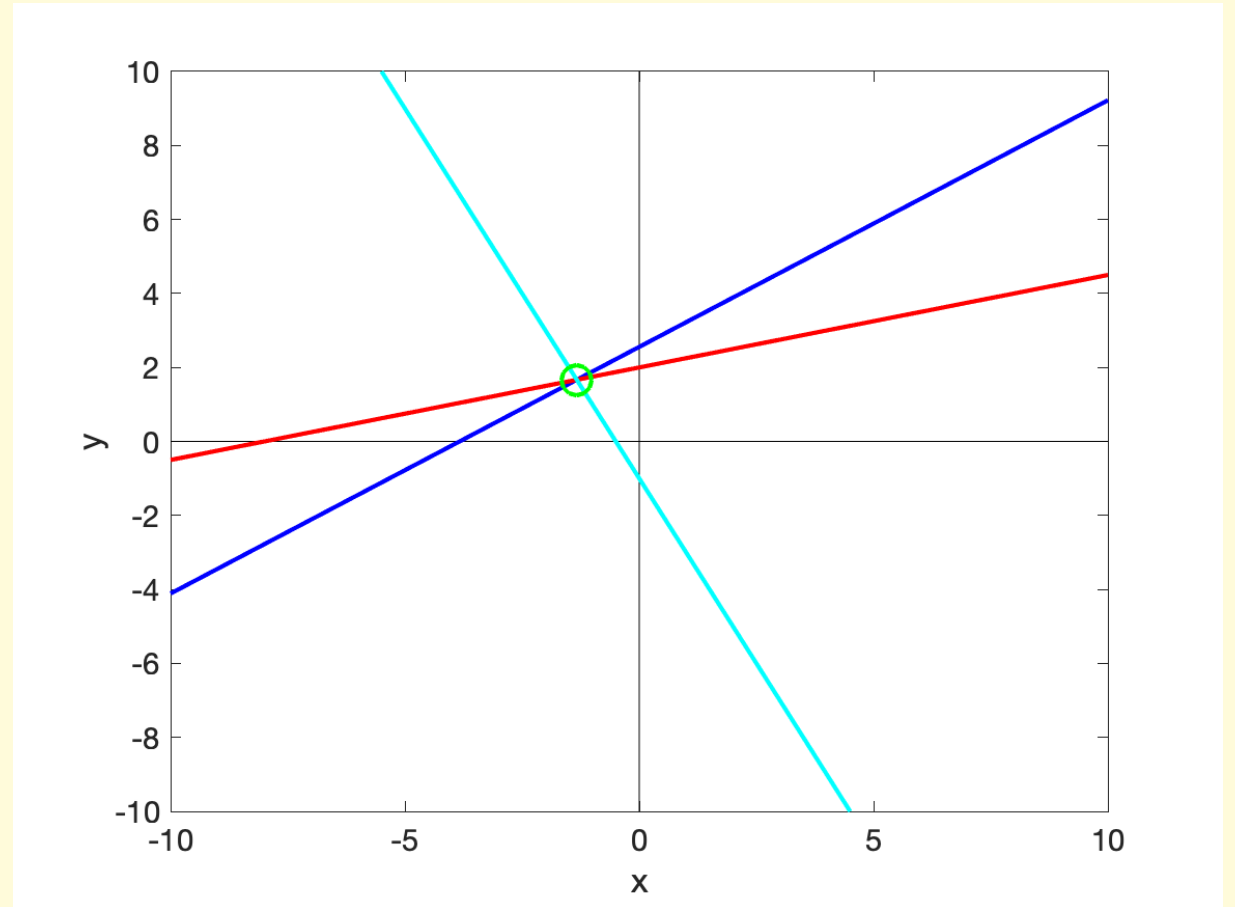
$$2x - 3y = 3 \rightarrow y = \frac{2}{3}x - \frac{23}{9}$$

$$x - 4y = 4 \rightarrow y = \frac{1}{4}x - 1$$

$$2x + y = -1 \rightarrow y = -2x - 1$$

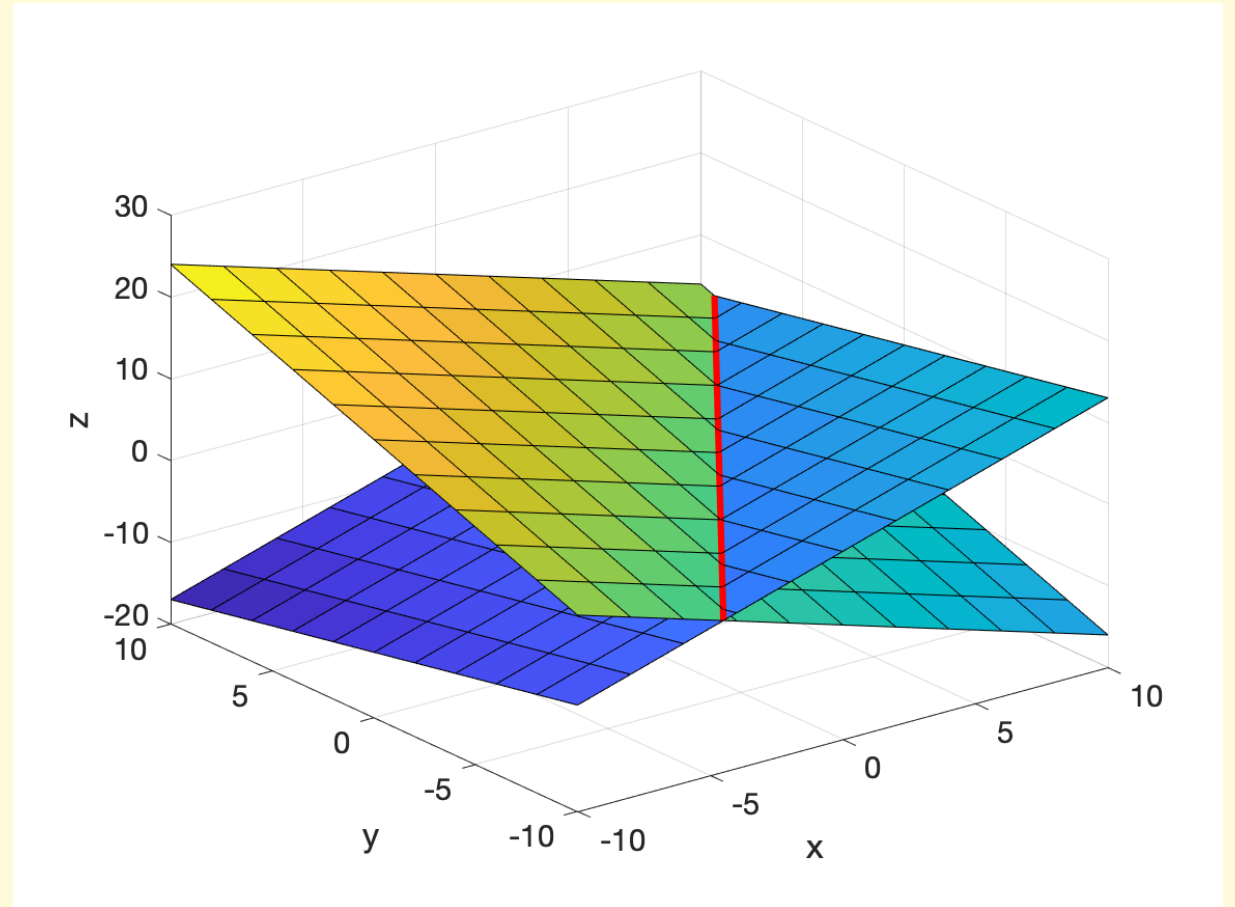
- The system has one unique solution.

Solution: $(x, y) = (-4/3, 5/3)$



The geometry of linear systems.

- The general form of a linear equation in 3-D is:
$$a_1x_1 + a_2x_2 + a_3x_2 = b$$
 - This is the equation for a plane with normal vector (a_1, a_2, a_3) .
- A system of linear equations in 3-D is thus a set of planes.
- Solutions to the system are points where the planes intersect.



The geometry of linear systems.

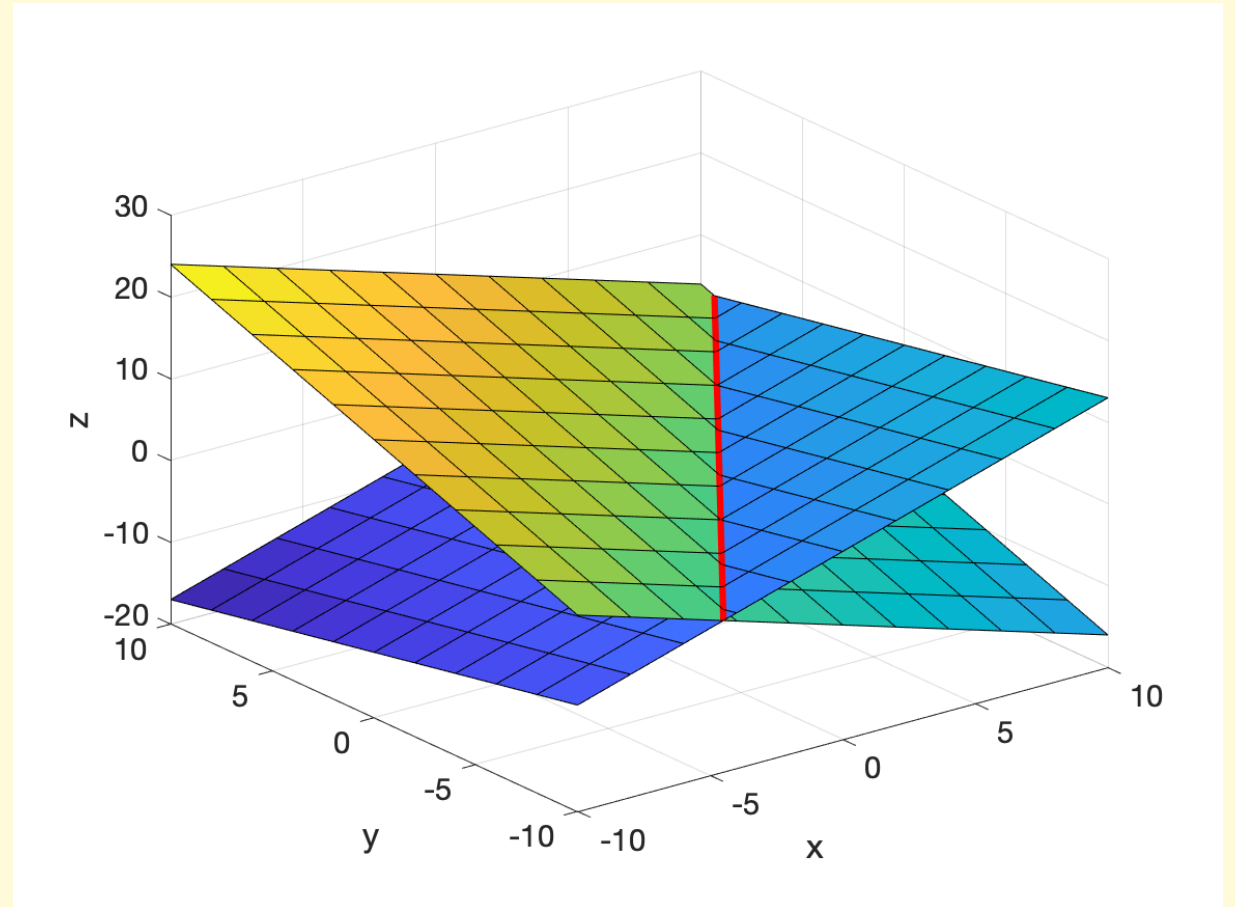
- A system of two planes in 3-D usually intersects in a line.

$$x - y + z = 4$$

$$x - \frac{y}{2} - z = 2$$

- The system has many solutions.
 - There are an infinite number of solutions that we can express in 'parametric' form.

Solution: $(x, y, z) = \left(\frac{3t}{4} + 3, t, \frac{t}{4} + 1\right), \forall t$



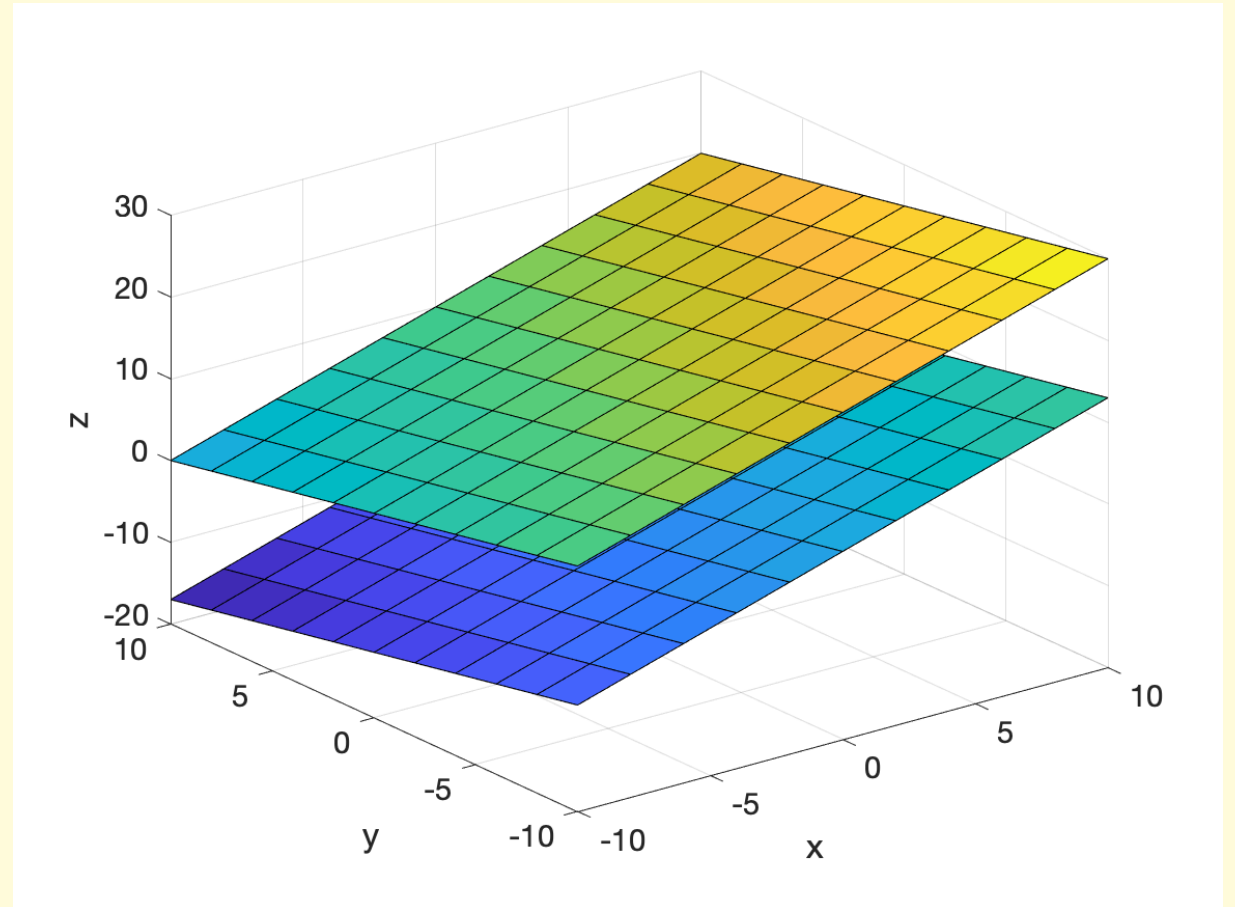
The geometry of linear systems.

- Parallel planes never intersect.

$$x - \frac{y}{2} - z = 2$$

$$x - \frac{y}{2} - z = -15$$

- **The system has no solution.**



The geometry of linear systems.

- A system of three planes in 3-D usually intersects at a single point.

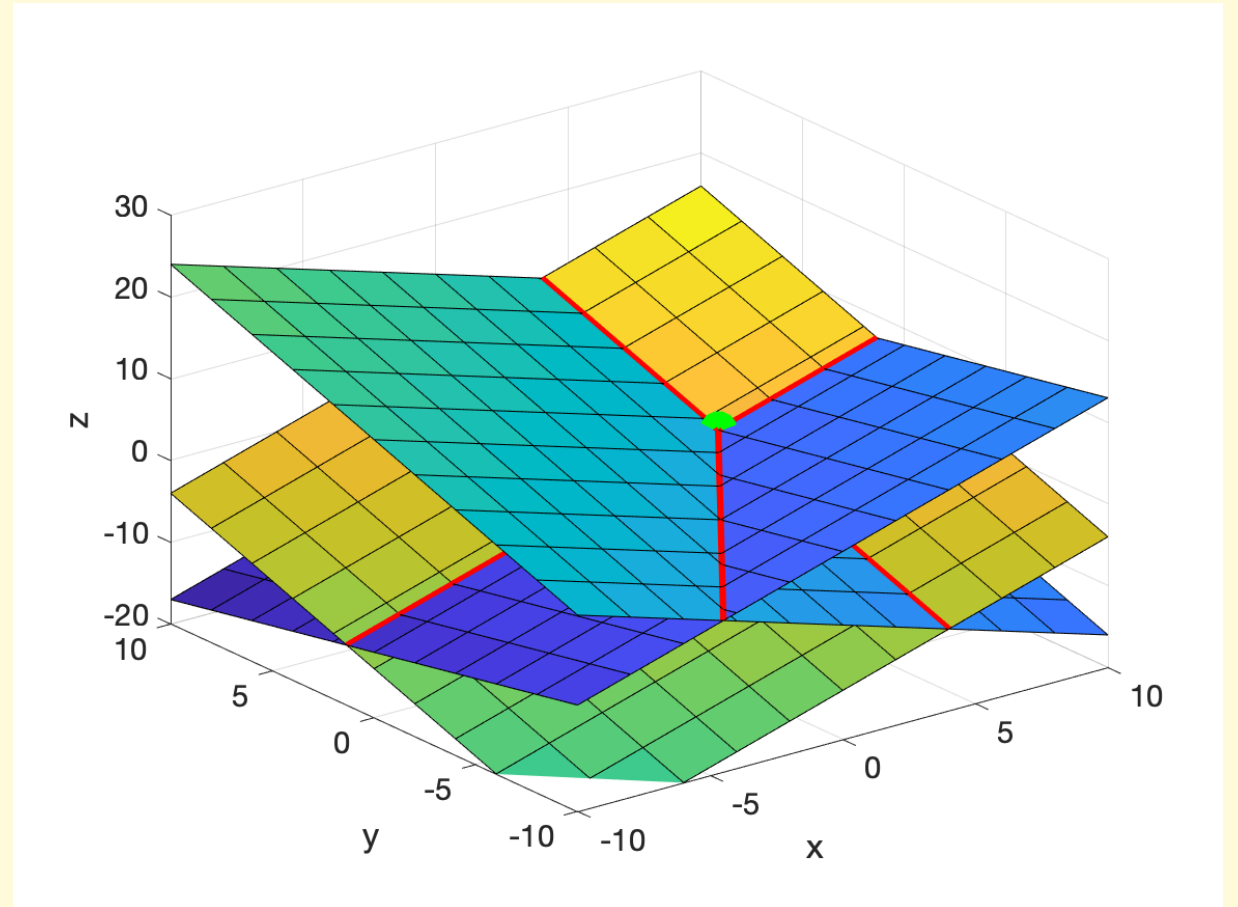
$$x - y + z = 4$$

$$x - \frac{y}{2} - z = 2$$

$$x + y - z = 4$$

- The system has one unique solution.

Solution: $(x, y, z) = (4, 4/3, 4/3)$



The geometry of linear equations.

- Systems of linear equations represent sets of “flat” geometric objects (lines in 2-D, planes in 3-D, hyperplanes in n -D) .
- Solutions to systems of linear equations are points where all the lines, planes, or hyperplanes intersect!
- **There may be a single intersection point, and thus a unique solution.**
- **There may be parallel objects, and thus no solution.**
- **There may be no common intersection point for all objects, and thus no solution.**
- **The objects may intersect along a line, on a plane, etc., and thus have an infinite number of solutions, represented in a parametric form.**
- **Linear systems can not have a finite number of solutions other than 0 or 1.**

An alternative view of linear systems.

- A system of equations can also be decomposed into a linear combination of vectors:

$$\begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x + 3y \\ x - y \end{bmatrix} = x \begin{bmatrix} 2 \\ 1 \end{bmatrix} + y \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

- The vectors in the linear combination are the columns of the coefficient matrix!
- Adding a right-hand side:

$$\begin{array}{l} 2x + 3y = 5 \\ x - 1y = -5 \end{array} \rightarrow x \begin{bmatrix} 2 \\ 1 \end{bmatrix} + y \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ -5 \end{bmatrix}$$

- Solutions to the system are the scaling factors needed to express the right-hand side as a linear combination of the columns of A

$$A\vec{x} = \vec{b}$$

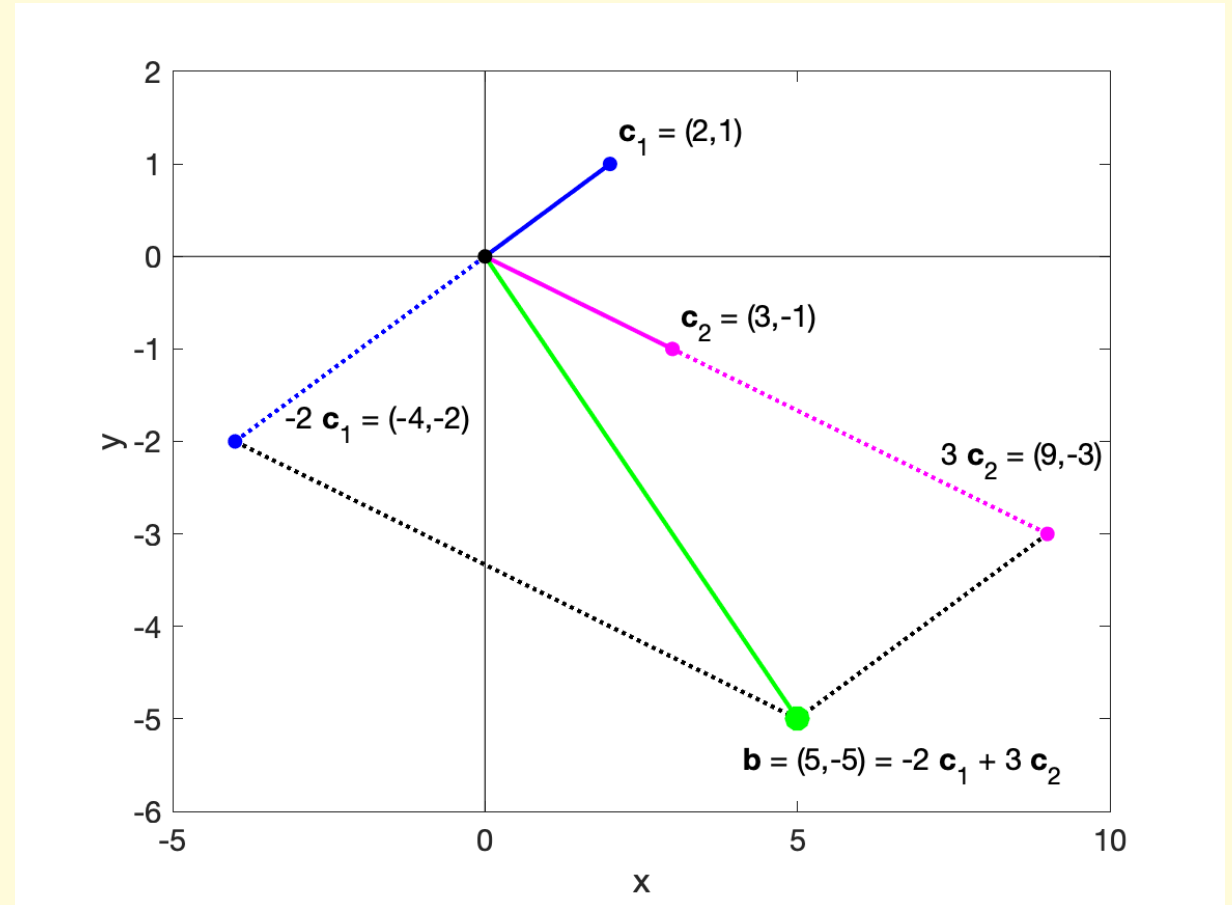
An alternative view of linear systems.

- Two equations in 2-D as a linear combination of vectors:

$$\begin{array}{l} 2x + 3y = 5 \\ x - 1y = -5 \end{array} \rightarrow x \begin{bmatrix} 2 \\ 1 \end{bmatrix} + y \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ -5 \end{bmatrix}$$

- Solutions to the system are the scaling factors needed to express the right-hand side as a linear combination of the columns of A:

$$\begin{array}{l} x = -2, y = 3 \\ -2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ -5 \end{bmatrix} \end{array}$$



An alternative view of linear systems.

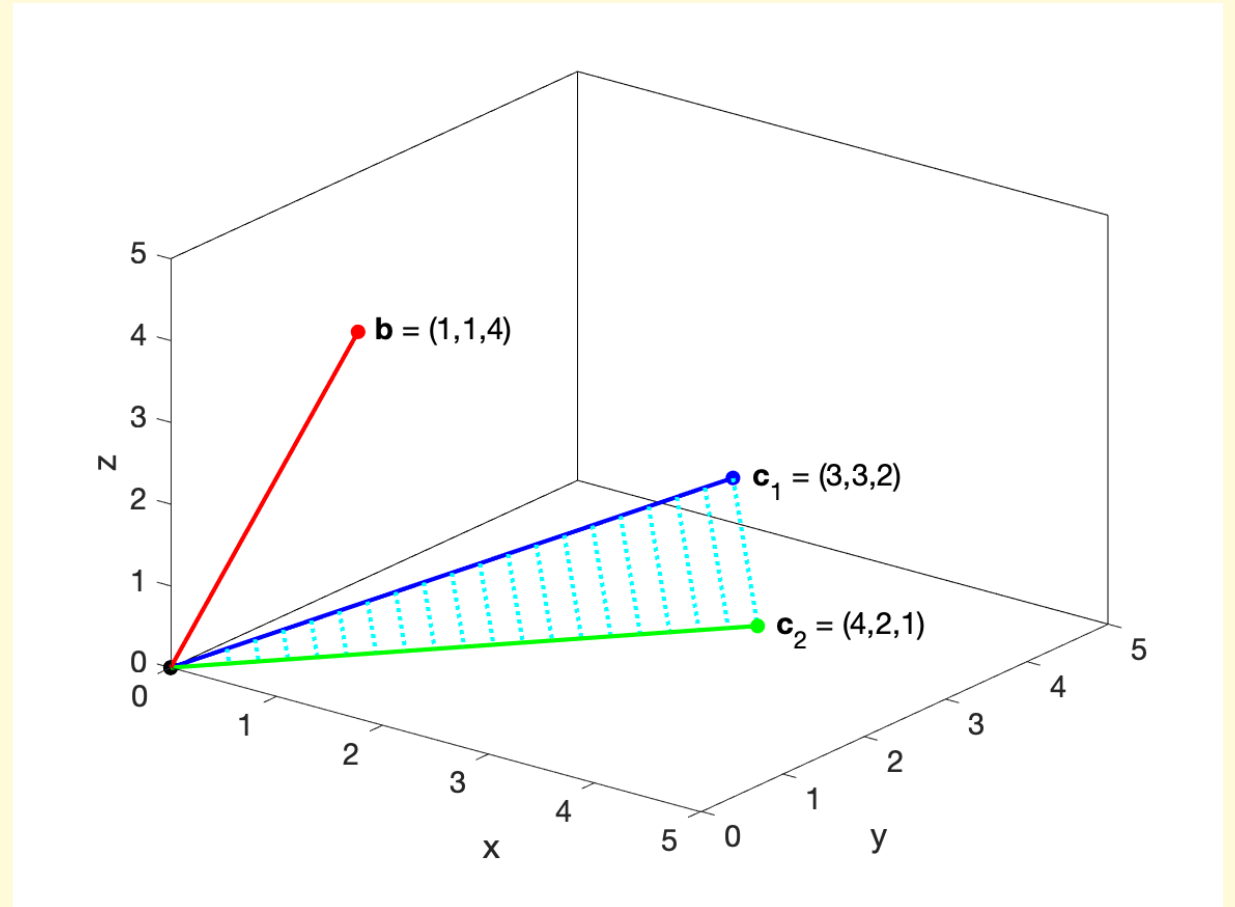
- When does a linear combination not exist?

$$\begin{array}{l} 3x + 4y = 1 \\ 3x + 2y = 1 \\ 2x + y = 4 \end{array} \rightarrow x \begin{bmatrix} 3 \\ 3 \\ 2 \end{bmatrix} + y \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix}$$

- The right-hand side vector does not lie in the same plane as the two column vectors!

$$\begin{array}{l} 3x + 4y = 1 \\ 3x + 2y = 1 \\ 2x + y = 2/3 + 0 \neq 4 \end{array} \rightarrow y = 0, x = 1/3$$

- There is no solution.



Two views of a linear system.

The row picture:

- Systems of linear equations represent sets of “flat” geometric objects (lines in 2-D, planes in 3-D, hyperplanes in n -D) .
- Solutions to systems of linear equations are points where **all** the lines, planes, or hyperplanes intersect.

The column picture:

- Systems of linear equations are an attempt to represent a given right-hand-side vector as a linear combination of column vectors on the left-hand side.
- Solutions to systems of linear equations are the coefficients of such a linear combination.

Unit 3.2

**Solving linear systems using
Gaussian elimination.**

Gaussian Elimination

- An algorithm to solve ANY system of linear equations, if there is a solution.
- Involves two key steps:
 - Forward elimination.
 - Back substitution.
- Both use the idea that subtracting a scalar multiple of one equation from another can eliminate a variable from that equation!

Forward Elimination

1. Work from top to bottom and left to right.
2. First, use the first entry in the first equation to eliminate (zero out) all entries below it (this entry is called a pivot).
3. Move to the second equation and repeat.
4. Move to the third equation and repeat.
5. ...
6. The result is an upper triangular form!

Forward Elimination – An example.

Original system:

$$\begin{array}{rrcr} x & +y & -z & = 1 \\ x & +3y & +z & = 1 \\ 2x & +3y & +3z & = -2 \end{array}$$

Step 1: Subtract 1st row from 2nd and twice the first row from the 3rd:

$$\begin{array}{rrcr} x & +y & -z & = 1 \\ & +2y & +2z & = 0 \\ & +y & +5z & = -4 \end{array}$$

1. Work from top to bottom and left to right.
2. First, use the first entry in the first equation to eliminate (zero out) all entries below it (this entry is called a pivot).
3. Move to the second equation and repeat.
4. Move to the third equation and repeat.
5. ...
6. The result is an upper triangular form!

Forward Elimination – An example.

After step 1:

$$\begin{array}{rrcr} x & +y & -z & = 1 \\ & +2y & +2z & = 0 \\ & +y & +5z & = -4 \end{array}$$

Step 2: Subtract half the 2nd row from the 3rd:

$$\begin{array}{rrcr} x & +y & -z & = 1 \\ & +2y & +2z & = 0 \\ & & +4z & = -4 \end{array}$$

1. Work from top to bottom and left to right.
2. First, use the first entry in the first equation to eliminate (zero out) all entries below it (this entry is called a pivot).
3. Move to the second equation and repeat.
4. Move to the third equation and repeat.
5. ...
6. The result is an upper triangular form!

Forward Elimination – An example.

After step 2:

$$\begin{array}{rrcr} x & +y & -z & = 1 \\ & +2y & +2z & = 0 \\ & & +4z & = -4 \end{array}$$

- Last row has a single variable, the second to last has only two, and the first has all three.
- **An upper triangular form!**
- VERY easy to solve.

1. Work from top to bottom and left to right.
2. First, use the first entry in the first equation to eliminate (zero out) all entries below it (this entry is called a pivot).
3. Move to the second equation and repeat.
4. Move to the third equation and repeat.
5. ...
6. The result is an upper triangular form!

Back Substitution

1. Work from bottom to top and right to left.
2. First, start with the last equation: The upper triangular form directly gives us the value of the last variable. Substitute this value in all of the equations above.
3. Move to the second-to-last non-zero row and repeat with the second-to-last variable.
4. Move to the third-to-last non-zero row and repeat with the third-to-last variable.
5. ...
6. **We have the solution!**

Back Substitution – An example.

After Forward Elimination:

$$\begin{array}{rrcr} x & +y & -z & = 1 \\ & +2y & +2z & = 0 \\ & & +4z & = -4 \end{array}$$

Step 1: Solve for z, and substitute answer in equations above.

$$\begin{array}{rrcr} x & +y & +1 & = 1 \\ & +2y & -2 & = 0 \\ & & +z & = -1 \end{array}$$

1. Work from bottom to top and right to left.
2. First, start with the last equation: The upper triangular form directly gives us the value of the last variable. Substitute this value in all of the equations above.
3. Move to the second-to-last non-zero row and repeat with the second-to-last variable.
4. Move to the third-to-last non-zero row and repeat with the third-to-last variable.
5. ...
6. We have the solution!

Back Substitution – An example.

After first substitution:

$$\begin{array}{rrcr} x & +y & +1 & = 1 \\ & +2y & -2 & = 0 \\ & & +z & = -1 \end{array}$$

Step 2: Solve for y, and substitute answer in equations above.

$$\begin{array}{rrcr} x & +1 & +1 & = 1 \\ & y & & = +1 \\ & & z & = -1 \end{array}$$

1. Work from bottom to top and right to left.
2. First, start with the last equation: The upper triangular form directly gives us the value of the last variable. Substitute this value in all of the equations above.
3. Move to the second-to-last non-zero row and repeat with the second-to-last variable.
4. Move to the third-to-last non-zero row and repeat with the third-to-last variable.
5. ...
6. We have the solution!

Back Substitution – An example.

After second substitution:

$$\begin{array}{rclcl} x & +1 & +1 & = & 1 \\ & y & & = & +1 \\ & & z & = & -1 \end{array}$$

Step 3: Solve for x and read off the final answer.

$$\begin{array}{rclcl} x & & & = & -1 \\ & y & & = & +1 \\ & & z & = & -1 \end{array}$$

• **Solution:** $(x, y, z) = (-1, 1, -1)$

1. Work from bottom to top and right to left.
2. First, start with the last equation: The upper triangular form directly gives us the value of the last variable. Substitute this value in all of the equations above.
3. Move to the second-to-last non-zero row and repeat with the second-to-last variable.
4. Move to the third-to-last non-zero row and repeat with the third-to-last variable.
5. ...
6. We have the solution!

Gaussian Elimination

- An algorithm to solve ANY system of linear equations, if there is a solution.
- In many cases, Gaussian Elimination proceeds smoothly giving a unique final solution.
- What happens when there is **no solution**?
- What happens when there are **many solutions**?
- Are there cases where a system **has a solution, and yet Gaussian elimination fail**?

Issues with elimination: **no solution.**

Original system:

$$\begin{array}{rrcr} x & +y & -z & = 0 \\ x & +2y & +2z & = 1 \\ 2x & +3y & +z & = -2 \end{array}$$

Step 1: Subtract 1st row from 2nd and twice the first row from the 3rd:

$$\begin{array}{rrcr} x & +y & -z & = 0 \\ & +y & +3z & = 1 \\ & +y & +3z & = -2 \end{array}$$

Warning sign: 2nd and 3rd row have the same LHS but different RHS!

Issues with elimination: no solution.

After step 1:

$$\begin{array}{rrcr} x & +y & -z & = 0 \\ & +y & +3z & = 1 \\ & +y & +3z & = -2 \end{array}$$

Step 2: Subtract the 2nd row from the 3rd:

$$\begin{array}{rrcr} x & +y & -z & = 0 \\ & +y & +3z & = 1 \\ & 0 & +0 & = -3 \end{array}$$

The last equation is $0=3$, an obviously false statement!

Gaussian Elimination (GE)

- GE is an algorithm to solve ANY system of linear equations, if there is a solution.
- These are conditions for non solution:
 1. If Forward Elimination results in an equation with 0 on the left-hand side, but a non-zero entry on the right, the system has no solution.
 2. If Forward Elimination results in two equations with the same left-hand sides, but different right-hand sides, the system has no solution.
- As soon as you have identified that a system has no solution, STOP.

Issues with elimination: many solutions.

Original system:

$$\begin{array}{rrcr} x & +y & -z & = 0 \\ 2x & +y & +z & = 1 \\ x & & +2z & = 1 \end{array}$$

Step 1: Subtract twice the 1st row from the 2nd and the 1st row from the 3rd:

$$\begin{array}{rrcr} x & +y & -z & = 0 \\ & -y & +3z & = 1 \\ & -y & +3z & = 1 \end{array}$$

Warning sign: 2nd and 3rd row are identical

Issues with elimination: many solutions.

After step 1:

$$\begin{array}{rrcr} x & +y & -z & = 0 \\ & -y & +3z & = 1 \\ & -y & +3z & = 1 \end{array}$$

Step 2: Subtract the 2nd row from the 3rd:

$$\begin{array}{rrcr} x & +y & -z & = 0 \\ & -y & +3z & = 1 \\ & 0 & +0 & = 0 \end{array}$$

The last equation is $0=0$; this is true, but trivially so.

Issues with elimination: many solutions.

After Forward Elimination:

$$x + y - z = 0$$

$$-y + 3z = 1$$

$$0 + 0 = 0$$

The last equation has no variables, and the second to last has two! How does back-substitution work?

Step 1: Solve for y in terms of z!

$$x + (3z - 1) - z = 0$$

$$y = 3z - 1$$

Issues with elimination: many solutions.

After first substitution:

$$\begin{array}{rclcl} x & + (3z - 1) & - z & = & 0 \\ & y & & = & 3z - 1 \end{array}$$

Step 2: Solve for x, in terms of z.

$$\begin{array}{rcl} x & = & 1 - 2z \\ y & = & 3z - 1 \end{array}$$

We can read off a solution of x and y in terms of z, but what is z?

z can be any value!

Issues with elimination: many solutions.

After second substitution:

$$\begin{array}{rcl} x & & = 1 - 2z \\ y & & = 3z - 1 \end{array}$$

Step 3: Assign a parameter to the “free variable,” z , and write the final solution:

Solution: $(x, y, z) = (1 - 2t, 3t - 1, t), \forall t$

Gaussian Elimination (GE)

- GE is an algorithm to solve ANY system of linear equations, if there is a solution.
- The condition for many solutions is:
 1. If Forward Elimination results in an equation with **0 on the left-hand side AND on the right**, ignore that equation and proceed with the process.
- If the upper triangular form contains free variables, then the system has many (infinite) solutions. In this case, **solve for the pivot variables in terms of the free variables, then write a parametric solution.**

Issues with elimination: row swaps needed.

Original system:

$$\begin{array}{rrcr} x & +y & -z & = 1 \\ x & +y & +z & = 2 \\ x & +2y & +2z & = 1 \end{array}$$

Step 1: Subtract the 1st row from the 2nd and from the 3rd:

$$\begin{array}{rrcr} x & +y & -z & = 1 \\ & & +2z & = 1 \\ & y & +3z & = 0 \end{array}$$

Problem: Second row is missing the pivot entry!

Issues with elimination: row swaps needed.

After step 1:

$$\begin{array}{rrcr} x & +y & -z & = 1 \\ & & +2z & = 1 \\ & y & +3z & = 0 \end{array}$$

Step 2: Swap rows 2 and 3

$$\begin{array}{rrcr} x & +y & -z & = 1 \\ & y & +3z & = 0 \\ & & +2z & = 1 \end{array}$$

Problem solved! We have an upper triangular form and may proceed with back-substitution.

Issues with elimination: row swaps needed.

After forward elimination:

$$\begin{array}{rrcr} x & +y & -z & = 1 \\ & y & +3z & = 0 \\ & & +2z & = 1 \end{array}$$

Substitution 1: Solve for z

$$\begin{array}{rrcr} x & +y & -1/2 & = 1 \\ & y & +3/2 & = 0 \\ & & z & = 1/2 \end{array}$$

Issues with elimination: row swaps needed.

After first substitution:

$$\begin{array}{rclcl} x & +y & -1/2 & = & 1 \\ & y & +3/2 & = & 0 \\ & & z & = & 1/2 \end{array}$$

Substitution 2: Solve for y

$$\begin{array}{rclcl} x & -3/2 & -1/2 & = & 1 \\ & y & & = & -3/2 \\ & & z & = & 1/2 \end{array}$$

Issues with elimination: row swaps needed.

After first substitution:

$$\begin{array}{rcl} x & -3/2 & -1/2 & = 1 \\ & y & & = -3/2 \\ & & z & = 1/2 \end{array}$$

Substitution 3: Solve for x

$$\begin{array}{rcl} x & & = 3 \\ & y & = -3/2 \\ & & z & = 1/2 \end{array}$$

Solution: $(x, y, z) = (3, -3/2, 1/2)$

Gaussian Elimination

- GE is an algorithm to solve ANY system of linear equations, if there is a solution.
- If during Forward Elimination, the pivot variable is missing from the row you need it in, **swap with a lower row** that contains it.
- The need to swap rows does not imply anything about the set of solutions.

Issues with elimination: missing pivot.

Original system:

$$x + y - z = 2$$

$$x + y + z = 0$$

$$x + y - 2z = 3$$

**Step 1: Subtract the 1st row from the 2nd
and from the 3rd:**

$$x + y - z = 2$$

$$+2z = -2$$

$$-z = 1$$

**Problem: Second row is missing the pivot entry, but
so is the third!**

Issues with elimination: missing pivot.

After step 1:

$$\begin{array}{rclcrcl} x & +y & -z & = & 2 \\ & & +2z & = & -2 \\ & & -z & = & 1 \end{array}$$

Step 2: Second column is a “free variable,” so skip, and move to the third.

Step 2a: Add half the 2nd row to the 3rd.

$$\begin{array}{rclcrcl} x & +y & -z & = & 2 \\ & & +2z & = & -2 \\ & & 0 & = & 0 \end{array}$$

Issues with elimination: missing pivot.

After forward elimination:

$$\begin{array}{cccc} x & +y & -z & = 2 \\ & & +2z & = -2 \end{array}$$

Substitution 1: Solve for z.

$$\begin{array}{cccc} x & +y & +1 & = 2 \\ & & z & = -1 \end{array}$$

Issues with elimination: missing pivot.

After forward elimination:

$$\begin{array}{rrcr} x & +y & -z & = 2 \\ & & +2z & = -2 \end{array}$$

Substitution 2: y is a free variable, solve for x in terms of y

$$\begin{array}{rcl} x & & = 1 - y \\ z & & = -1 \end{array}$$

Issues with elimination: missing pivot.

After second substitution:

$$\begin{aligned}x &= 1 - y \\ z &= -1\end{aligned}$$

Step 3: Assign a parameter to the free variable, y , and write the final solution:

Solution: $(x, y, z) = (1 - t, t, -1), \forall t$

Gaussian Elimination (GE)

- GE is an algorithm to solve ANY system of linear equations, if there is a solution.
- If during Forward Elimination, the **pivot variable is missing** from the row you need it in, and from all lower rows, it is a free variable, and the system has multiple solutions.
- Shift right-ward to the **next pivot** and continue elimination, then back substitution.
- Solve for the pivot variables in terms of the free variables, then write a **parametric solution**.

Gaussian Elimination (GE)

GE is an **algorithm** to solve ANY system of linear equations, if there is a solution, and to identify when there is no solution.

- When there is a unique solution, forward elimination followed by back-substitution directly gives that solution.
- If there is no solution, a row of with a zero left-hand side and non-zero right-hand side will appear during forward elimination.
- If there are many solutions, at least one variable will not be a pivot; these are the free variables. Back-substitution can give a solution for the pivots in terms of the free variables, and this general solution can be written in parametric form.

Unit 3.2.1

**Worked problems on solving
linear systems with elimination.**

Solve the system of equations.

Find the solution to the following system or demonstrate that there is no solution.

$$\begin{array}{rrcr} -2x & +2y & -z & = 1 \\ x & +y & -z & = 2 \\ 3x & +2y & +2z & = 3 \end{array}$$

Solve the system of equations.

$$-2x + 2y - z = 1$$

$$x + y - z = 2$$

$$3x + 2y + 2z = 3$$

- Add $\frac{1}{2}$ Eq. 1 to Eq. 2, and $\frac{3}{2}$ Eq. 1 to Eq. 3

$$-2x + 2y - z = 1$$

$$0 + 2y - \left(\frac{3}{2}\right)z = \frac{5}{2}$$

$$0 + 5y + \left(\frac{1}{2}\right)z = \frac{9}{2}$$

Solve the system of equations.

$$\begin{array}{rclcl} -2x & +2y & -z & & = 1 \\ & +2y & -\left(\frac{3}{2}\right)z & & = \frac{5}{2} \\ & +5y & +\left(\frac{1}{2}\right)z & & = \frac{9}{2} \end{array}$$

- Subtract $\frac{5}{2}$ Eq. 2 from Eq. 3

$$\begin{array}{rclcl} -2x & +2y & -z & & = 1 \\ & +2y & -\left(\frac{3}{2}\right)z & & = \frac{5}{2} \\ & 0 & +\left(\frac{1}{2} + \frac{15}{4}\right)z & & = \frac{9}{2} - \frac{25}{4} \end{array}$$

Solve the system of equations.

$$\begin{aligned} -2x + 2y - z &= 1 \\ +2y - \left(\frac{3}{2}\right)z &= \frac{5}{2} \\ +\left(\frac{17}{4}\right)z &= -\frac{7}{4} \end{aligned}$$

- Solve for z, and substitute

$$\begin{aligned} -2x + 2y + \left(\frac{7}{17}\right) &= 1 \\ +2y - \left(\frac{3}{2}\right)\left(\frac{-7}{17}\right) &= \frac{5}{2} \\ +z &= \frac{-7}{17} \end{aligned}$$

Solve the system of equations.

$$\begin{aligned} -2x + 2y + \left(\frac{7}{17}\right) &= 1 \\ +2y + \left(\frac{21}{34}\right) &= \frac{5}{2} \\ +z &= \frac{-7}{17} \end{aligned}$$

- Solve for y, and substitute

$$\begin{aligned} -2x + 2\left(\frac{16}{17}\right) + \left(\frac{7}{17}\right) &= 1 \\ +y &= \frac{1}{2}\left(\frac{5}{2} - \frac{21}{34}\right) = \frac{16}{17} \\ +z &= \frac{-7}{17} \end{aligned}$$

Solve the system of equations.

$$\begin{array}{rclcl} -2x & + & \left(\frac{32}{17}\right) & + & \left(\frac{7}{19}\right) & = & 1 \\ & & +y & & & = & \frac{16}{17} \\ & & & & +z & = & \frac{-7}{17} \end{array}$$

- Solve for x

$$\begin{array}{rclcl} x & = & \frac{-1}{2} \left(1 - \frac{32}{17} - \frac{7}{17} \right) & = & \frac{11}{17} \\ & & +y & = & \frac{16}{17} \\ & & & & +z & = & \frac{-7}{17} \end{array}$$

Solve the system of equations.

$$\begin{array}{rcl} x & & = \frac{11}{17} \\ & y & = \frac{16}{17} \\ & & z = \frac{-7}{17} \end{array}$$

- Final solution:

$$(x, y, z) = \left(\frac{11}{17}, \frac{16}{17}, \frac{-7}{17} \right) = \frac{1}{17} (11, 16, -7)$$

Solve the system of equations.

Find the solution to the following system or demonstrate that there is no solution.

$$\begin{array}{rrcr} 3x & +2y & -2z & = 1 \\ x & +y & -2z & = 2 \\ 2x & +y & & = 0 \end{array}$$

Solve the system of equations.

$$\begin{array}{rrcr} 3x & +2y & -2z & = 1 \\ x & +y & -2z & = 2 \\ 2x & +y & & = 0 \end{array}$$

- Subtract $\frac{1}{3}$ Eq. 1 from Eq. 2, and $\frac{2}{3}$ Eq. 1 from Eq. 3

$$\begin{array}{rrcr} 3x & +2y & -2z & = 1 \\ 0 & +\left(\frac{1}{3}\right)y & -\left(\frac{4}{3}\right)z & = \frac{5}{3} \\ 0 & -\left(\frac{1}{3}\right)y & +\left(\frac{4}{3}\right)z & = \frac{-2}{3} \end{array}$$

Solve the system of equations.

$$\begin{array}{rclcrcl} 3x & +2y & -2z & = & 1 \\ & +\left(\frac{1}{3}\right)y & -\left(\frac{4}{3}\right)z & = & \frac{5}{3} \\ & -\left(\frac{1}{3}\right)y & +\left(\frac{4}{3}\right)z & = & -\frac{2}{3} \end{array}$$

- Add Eq. 2 to Eq. 3

$$\begin{array}{rclcrcl} 3x & +2y & -2z & = & 1 \\ & +\left(\frac{1}{3}\right)y & -\left(\frac{4}{3}\right)z & = & \frac{5}{3} \\ & 0 & 0 & = & 1 \end{array}$$

- Eq. 3 is inconsistent: This system has no solution.

Solve the system of equations.

Find the solution to the following system or demonstrate that there is no solution.

$$\begin{array}{rclcrcl} & +2y & -2z & = & -1 \\ x & +y & +2z & = & 2 \\ x & +3y & & = & 1 \end{array}$$

Solve the system of equations.

$$\begin{array}{rrcr} & +2y & -2z & = -1 \\ x & +y & +2z & = 2 \\ x & +3y & & = 1 \end{array}$$

- Swap Eq. 1 and Eq. 2

$$\begin{array}{rrcr} x & +y & +2z & = 2 \\ & +2y & -2z & = -1 \\ x & +3y & & = 1 \end{array}$$

- Subtract Eq. 1 from Eq. 3

$$\begin{array}{rrcr} x & +y & +2z & = 2 \\ & +2y & -2z & = -1 \\ 0 & +2y & -2z & = -1 \end{array}$$

Solve the system of equations.

$$\begin{array}{rrcr} x & +y & +2z & = 2 \\ & +2y & -2z & = -1 \\ & +2y & -2z & = -1 \end{array}$$

- Subtract Eq. 2 from Eq. 3

$$\begin{array}{rrcr} x & +y & +2z & = 2 \\ & +2y & -2z & = -1 \\ & 0 & 0 & = 0 \end{array}$$

- There will be multiple solutions; z is the free variable.
- Solve for pivots (x and y) in terms of z .

Solve the system of equations.

$$\begin{array}{rclcl} x & +y & +2z & = & 2 \\ & +2y & -2z & = & -1 \end{array}$$

- Solve for y, and substitute:

$$\begin{array}{rclcl} x & +y & +2z & = & 2 \\ & y & & = & z - \frac{1}{2} \end{array}$$

- Substitute:

$$\begin{array}{rclcl} x & + \left(z - \frac{1}{2} \right) & +2z & = & 2 \\ & y & & = & z - \frac{1}{2} \end{array}$$

Solve the system of equations.

$$\begin{aligned}x + \left(z - \frac{1}{2}\right) + 2z &= 2 \\ y &= z - \frac{1}{2}\end{aligned}$$

- Solve for x, and substitute:

$$\begin{aligned}x &= -3z + \frac{5}{2} \\ y &= z - \frac{1}{2}\end{aligned}$$

- Write in parametric form, with $z=t$:

$$(x, y, z) = \left(-3t + \frac{5}{2}, t - \frac{1}{2}, t\right), \forall t \in \mathbb{R}$$